



Topic

1. Introduction to Basic Concepts of JavaScript
2. Data Types, Operators, and Functions in JavaScript
3. JavaScript in HTML

Objectives

Students are expected to:

1. Understanding the concept of Javascript
2. Understanding Data types, operators and functions in javascript
3. Students are able to run javascript in HTML files

Introduction

JavaScript is a client-side programming language used for web development. A client-side programming language means that the processing is done on the client's side. The client application in this case refers to web browsers such as Google Chrome and Mozilla Firefox. Client-side programming languages differ from server-side programming languages like PHP, where all the program code is executed on the server side.

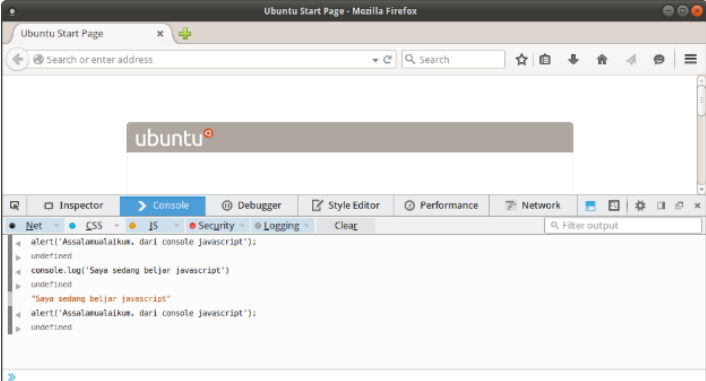
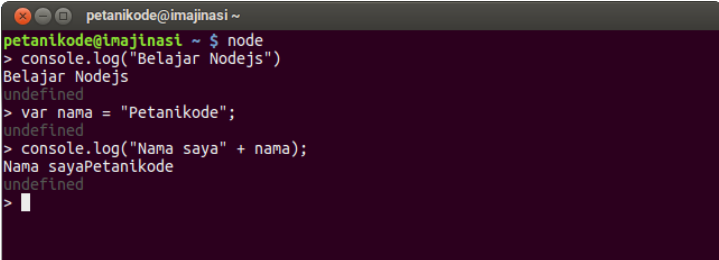
To run JavaScript, you only need a text editor and a web browser. JavaScript offers features such as being a high-level programming language, client-side, loosely typed, and object-oriented. Initially, JavaScript was developed to make interactions between users and websites faster without having to wait for processing on the web server. Before JavaScript, every interaction from the user had to be processed by the web server.

Imagine when you fill out a registration form on a website, click the submit button, wait about 10 seconds for the website to process the form, and then receive a page stating that some form fields were left unfilled. This is the kind of situation for which JavaScript was developed. The processing to check whether all form fields have been filled can be transferred from the web server to the web browser.

As JavaScript evolved, it became useful not only for form validation but also for many modern purposes. Various animations to beautify web pages, chat features, modern effects, games—all of these can be created using JavaScript. There are 3 main ways to write JavaScript tags:

1. Writing the tag with `<script type="text/javascript">` at the start and ending with `</script>`. The attribute informs the browser that the script within the tag is JavaScript in text format.
2. Writing the tag with `<script language="javascript">` at the start and ending with `</script>`. This attribute is used to specify the version of JavaScript being used. For example, `<script language="javascript1.2">` indicates that the version of JavaScript used is 1.2.
3. Writing the tag with `<script language="javascript" type="text/javascript">` at the start and ending with `</script>`. This mixed method combines the old and new ways of writing, allowing compatibility for web browsers that support JavaScript but may not yet support HTML fully.

Practical Section 1: Learning Javascript

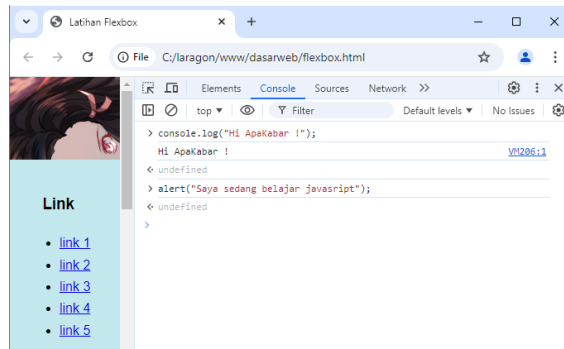
Step	Description
1	We can open the JavaScript console through Inspect Element -> Console.  In the console, we can write functions or JavaScript code, and the results will be displayed immediately.
2	For example, let's try the following code: <pre>console.log("Hi apa kabar!"); alert("Saya sedang belajar javascript");</pre> Observe what appears on the console, then record your observations!.
3	If you are using Nodejs, then the way to access <i>the console</i> is to type the node's command in the Terminal. 

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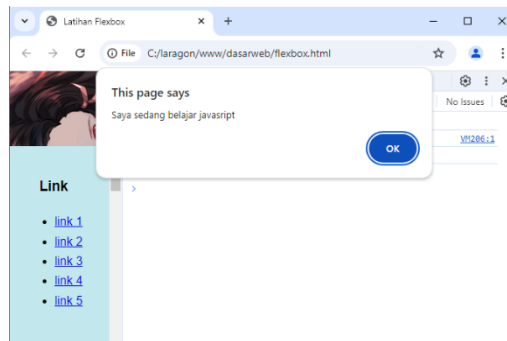
Observe what happens, then record your observations. What can be concluded after trying *the Javascript* console?

(Question No.1)

1. Input

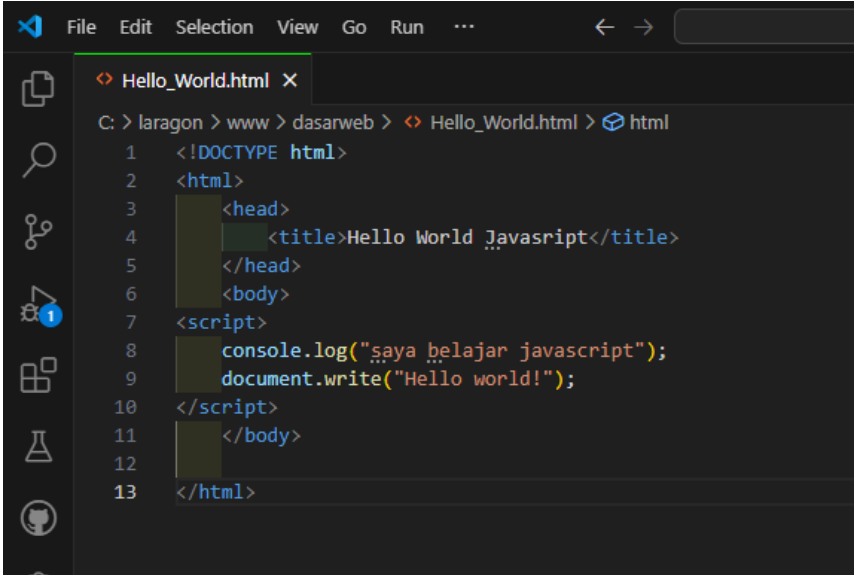


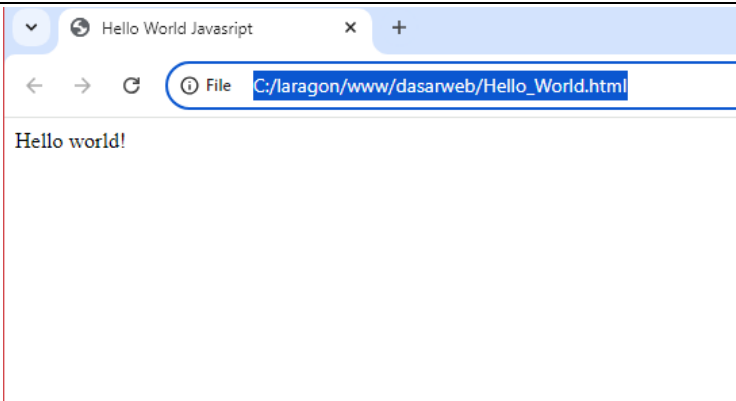
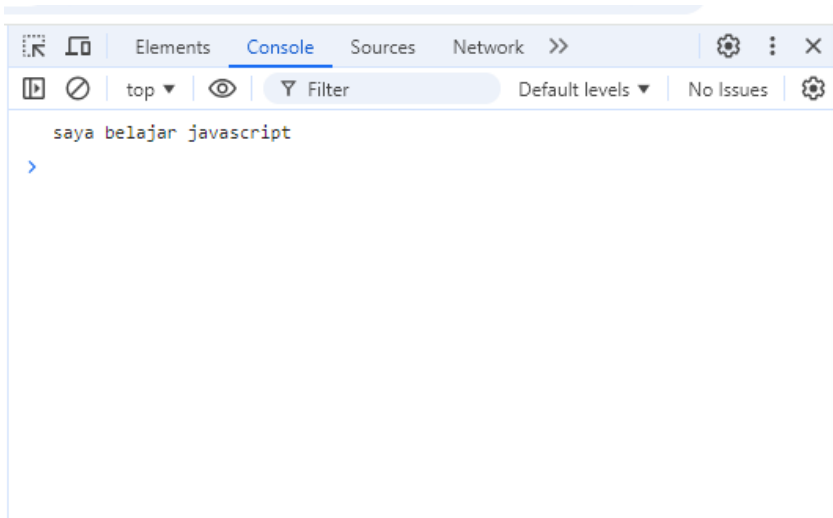
2. Output

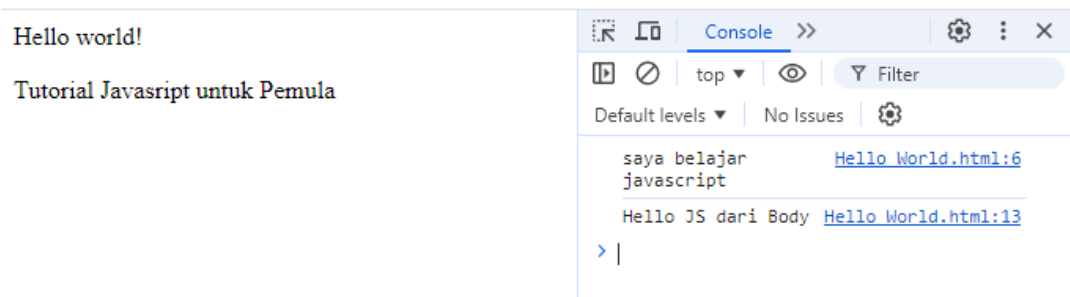


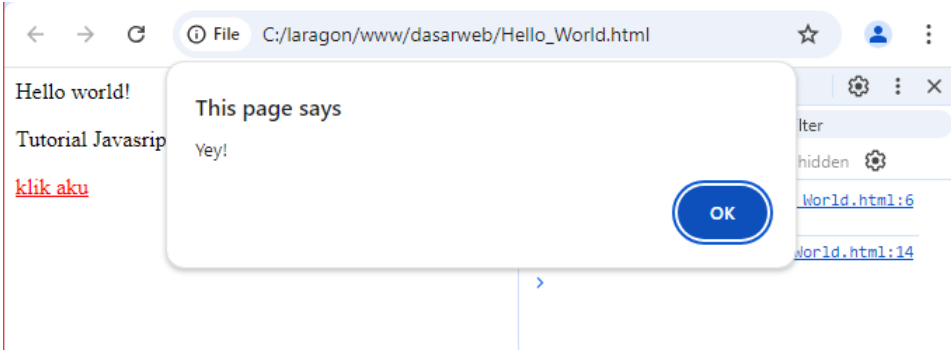
Practical Section 2: Creating the First Javascript Program

Step	Description
1	Please open a text editor, then create a new file named <code>hello_world.html</code>
2	Type the program code below:

	<pre> <!DOCTYPE html> <html> <head> <title>Hello World Javascript</title> </head> <body> <script> console.log("Saya belajar Javascript"); document.write("Hello World!"); </script> </body> </html> </pre>
3	Save it as hello_world.html , then open the file with a web browser.
4	Observe what happens in the browser, then record your observations (Question No.2) - Input  - Output

	
5	Now try to open the javascript console, right click page in the browser, then choose Inspect Elements > Console
6	Observe what happens in the Console tab, then record your results! (Question No. 3) 
7	Earlier, we wrote the command: <pre>console.log("Saya belajar Javascript");</pre> Why do you think the command is not displayed? (Question No.4)

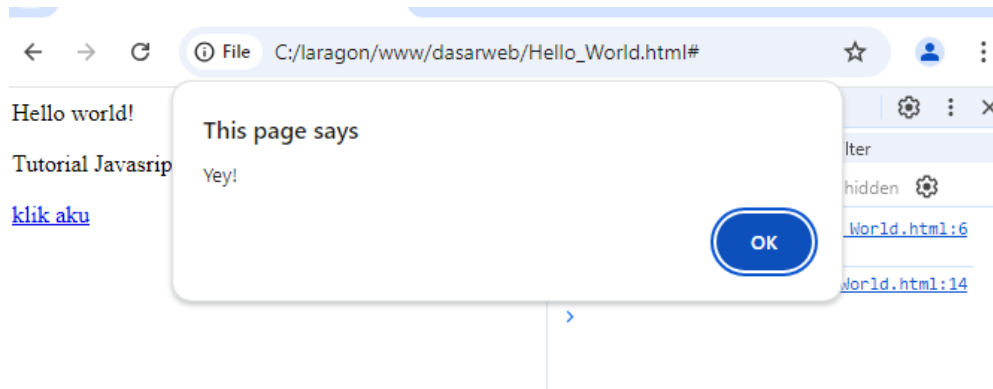
	<pre> <!DOCTYPE html> <html> <head> <title>Belajar Javascript dari Nol</title> <script> // ini adalah penulisan kode javascript // di dalam tag <head> console.log("Hello JS dari Head"); </script> </head> <body> <p>Tutorial Javascript untuk Pemula</p> <script> // ini adalah penulisan kode javascript // di dalam tag <body> console.log("Hello JS dari body"); </script> </body> </html> </pre>
3	Observe what happens to the browser? Record your observations (Question No. 5) 
4	Which do you think is better, written in the <code><head></code> or <code><body></code> tag? (Question No. 6) placing JavaScript in the <code><body></code> (especially at the end of the <code><body></code> tag) is better. Because: Improved Page Load Speed: - The HTML content loads and displays first, so users can see the page before the JavaScript starts running. This makes the page appear faster, enhancing user experience. Availability of DOM Elements: - Since the entire page's HTML content is already loaded, JavaScript can safely interact with any element on the page. This is especially important if your script needs to modify or interact with the DOM.

2. Inline Javascript Code Writing	
Step	Description
1	In this way, we'll write the javascript code inside the HTML attribute. This method is usually used to call a function on a specific event. One example is when clicked.
2	Type the program code below: <pre>Klik aku!</pre> Or it can also be like this: <pre>Klik aku!</pre>
3	Observe what happens to the browser! Record your observations (Question No. 7) a. Input1 <pre> 1 <!DOCTYPE html> 2 <html> 3 <head> 4 <title>Belajar Javascript dari Nol</title> 5 <script> 6 console.log("saya belajar javascript"); 7 document.write("Hello world!"); 8 </script> 9 </head> 10 <body> 11 <p>Tutorial Javascrpt untuk Pemula</p> 12 klik aku 13 <script> 14 console.log("Hello JS dari Body") 15 </script> 16 </body> 17 18 </html></pre> b. Output1 

c. Input2

```
<? Hello_World.html >
C: > laragon > www > dasarweb > <? Hello_World.html >
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Belajar Javascript dari Nol</title>
5          <script>
6              console.log("saya belajar javascript");
7              document.write("Hello world!");
8          </script>
9      </head>
10     <body>
11         <p>Tutorial Javascrript untuk Pemula</p>
12         <a href="javascript:alert('Yey!')">klik aku</a>
13     </script>
14         console.log("Hello JS dari Body")
15     </script>
16 </body>
17
18 </html>
```

d. Output2



4

What is the difference between the two program codes

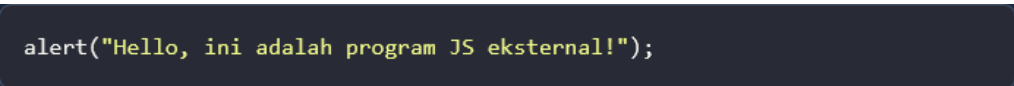
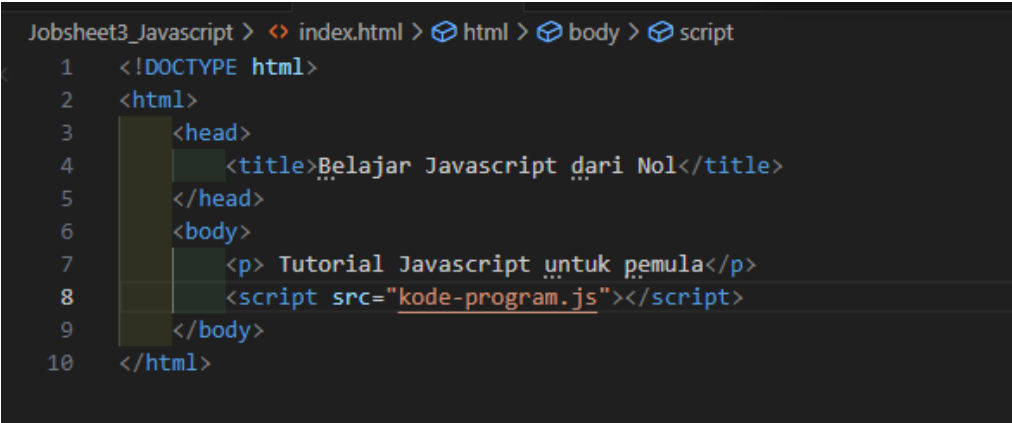
(Question No. 8)

The main differences:

- **Code 1:** JavaScript is executed directly through `href="javascript:"`, which can cause the page to reload or lead to unexpected behavior.
- **Code 2:** JavaScript is executed through `onclick` with `href="#"`, which by default will scroll to the top of the page but can be adjusted.

Conclusion: Code 2 is better because it's cleaner and separates HTML from JavaScript.

3. External Javascript Code Writing

Step	Description
1	In this way, we'll write the javascript code separately from the HTML file. In this method, we will write JavaScript code separately from the HTML file. This approach is usually recommended for large projects, as it makes project code management easier.
2	Let's try, create two files, namely HTML and Javascript files.  <pre> belajar-js/ ├── kode-program.js └── index.html </pre>
3	Contents of the kode-program.js file :  <pre> alert("Hello, ini adalah program JS eksternal!"); </pre>
4	Contents of the index.html file :  <pre> <!DOCTYPE html> <html> <head> <title>Belajar Javascript dari Nol</title> </head> <body> <p>Tutorial Javascript untuk Pemula</p> <!-- Menyisipkan kode js eksternal --> <script src="kode-program.js"></script> </body> </html> </pre>
5	Observe what happens to the browser! Record your observations (Question No. 9) 1. Input  <pre> Jobsheet3_Javascript > index.html > html > body > script 1 <!DOCTYPE html> 2 <html> 3 <head> 4 <title>Belajar Javascript dari Nol</title> 5 </head> 6 <body> 7 <p> Tutorial Javascript untuk pemula</p> 8 <script src="kode-program.js"></script> 9 </body> 10 </html> </pre>

	2. Output
6	In the experiment, we wrote separate javascript code with HTML code. Then in the HTML code we insert the <code>src</code> attribute in the <code><script></code> tag <pre> <!-- Menyesipkan kode js eksternal --> <script src="kode-program.js"></script> </pre> Then anything in <code>kode-program.js</code> file will be readable from <code>index.html</code> file
7	What would happen if the javascript file was in a different folder? Observe and record your observations (Question No. 10) If the JavaScript file is in a different folder from the HTML file, you need to ensure the correct path is specified in the <code>src</code> attribute of the <code><script></code> tag. If the path is correct, the JavaScript file will be successfully loaded and executed by the browser, allowing the JavaScript code to function as intended. However, if the path is incorrect or the file does not exist at the specified location, the browser will display a "404 Not Found" error in the console, indicating that the file was not found and the JavaScript code will not run.
8	Suppose we have a folder structure like this: <pre> belajar-js/ ├── js/ │ └── kode-program.js └── index.html </pre> So to insert the <code>kode-program.js</code> file into the HTML, we can write the following code:

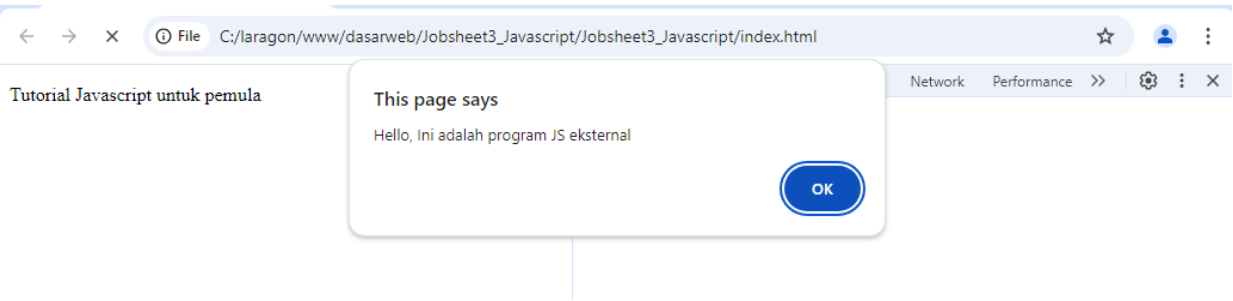
```
<script src="js/kode-program.js"></script>
```

Because the `kode-program.js` file is in the `js` directory.

We can also insert javascript that exists on the internet by providing the full URL address.

Example:

```
<script src="https://www.petanikode.com/js/kode.js"></script>
```



Practical Section 4: Dialogue Window

A dialog window is a window used to interact with users. There are three types of dialog windows in Javascript:

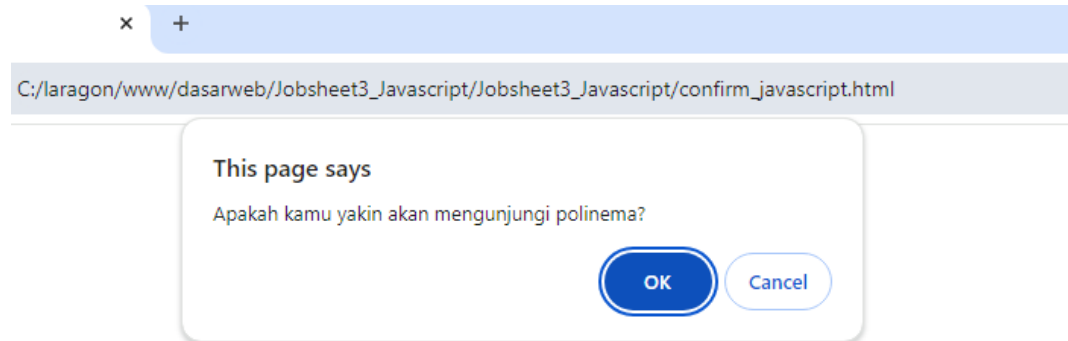
1. The `alert()` dialog window;
2. The `confirm()` dialog window;
3. The prompt dialog `window()` ;

Step	Description
1	Create a new file <code>alert_javascript.html</code> and save it in the project folder.
2	Type the program code below
	<pre><html> <head> <script type="text/javascript"> function message() { alert("This alert box was called with the onload event") } </script> </head> <body onload="message()"> </body> </html></pre>
3	Observe what appears on the browser
4	Record your observations (Question No. 11) 1. Input

	<pre> <> alert_javascript.html x <> alert_javascript.html > html > body 1 <html> 2 <head> 3 <script type="text/Javascript"> function message(){ 4 alert("This Alert box was called With the onload event");}</scrip 5 </head> 6 <body onload="message()"></body> 7 </html> </pre> 2. Output
5	Create a new file named confirm_javascript.html and save it in the project folder
6	Type the program code below <pre> <!DOCTYPE html> <html> <head> <title>Dialog Confirm</title> </head> <body> <script> var yakin = confirm("Apakah kamu yakin akan mengunjungi polinema?"); if (yakin) { window.location = "https://www.polinema.ac.id"; } else { document.write("Baiklah, tetap di sini saja :)"); } </script> </body> </html> </pre>
7	Observe what appears on the browser
8	Record your observations (Question No. 12) 1. Input

```
confirm_javascript.html x
confirm_javascript.html > html
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Dialog Confirm</title>
5  </head>
6  <body>
7    <script>
8      var yakin = confirm("Apakah kamu yakin akan mengunjungi polinema?")
9      if (yakin) {
10        window.location = "https://www.polinema.ac.id";
11      } else {
12        document.write("Baiklah, tetap disini saja yaaa :)");
13      }
14    </script>
15  </body>
16 </html>
```

2. Output



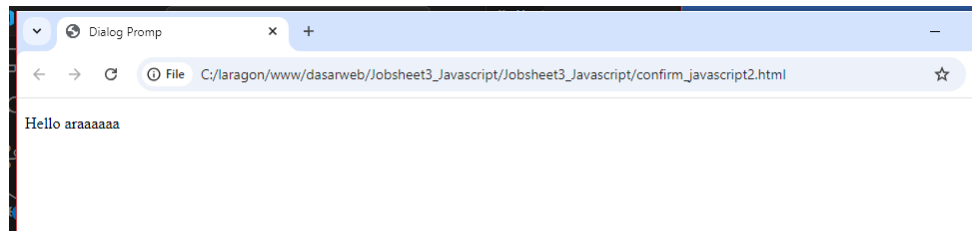
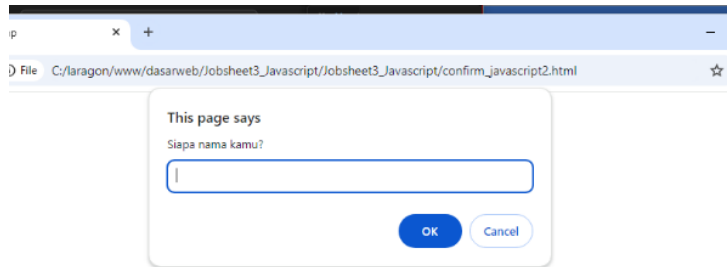
	
9	Type the program code below <pre data-bbox="235 945 1079 1344"> <!DOCTYPE html> <html> <head> <title>Dialog Promp</title> </head> <body> <script> var nama = prompt("Siapa nama kamu?", ""); document.write("<p>Hello "+ nama +"</p>"); </script> </body> </html> </pre>
10	Observe what appears on the browser
11	Record your observations (Question No. 13) 1. Input

```

<> confirm_javascript2.html > html
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Dialog Prompt</title>
5      </head>
6      <body>
7          <script>
8              var nama = prompt("Siapa nama kamu?", "");
9              document.write("<p>Hello " + nama + "</p>");
10         </script>
11     </body>
12 </html>

```

2. Output



Practical Section 5: Variables

The way to create a variable that is commonly used in javascript is to use the var keyword followed by the name of the variable and its value.

Example: `var title = "Learn Javascript Programming";`

Displaying the contents of a Variable

To display the contents of the variables, we can utilize functions to display outputs such as:

- The `console.log()` function returns the output to the javascript console;
- The `document.write()` function returns the output to an HTML document;
- and the `alert()` function returns the output to the dialog window.

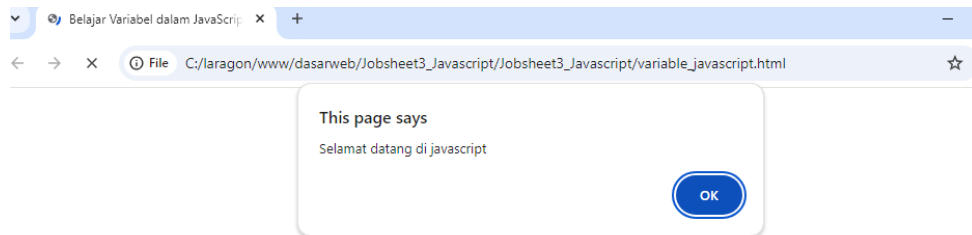
Step	Description
1	Create a new file <code>variable_javascript.html</code> and save it in the project folder.
2	Type the program code below
	<pre><!DOCTYPE html> <html lang="en"> <head> <title>Belajar Variabel dalam Javascript</title> <script> // membuat variabel var name = "Javascript"; var visitorCount = 50322; var isActive = true; // menampilkan variabel ke jendela dialog (alert) alert("Selamat datang di " + name); // menampilkan variabel ke dalam HTML document.write("Nama Situs: " + name + "
"); document.write("Jumlah Pengunjung: " + visitorCount + "
"); document.write("Status Aktif: " + isActive + "
"); </script> </head> <body> </body> </html></pre>
3	Observe what appears on the browser
4	Record your observations (Question No. 14) 1. input

```

<> variable_javascript.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3      <head>
4          <title>Belajar Variabel dalam JavaScript</title>
5          <script>
6              var Name = "javascript";
7              var VisitorCount = 50322;
8              var IsActive = true;
9              alert("Selamat datang di " + Name);
10             document.write("Nama Situs: " + Name + "<br>");
11             document.write("Jumlah pengunjung: " + VisitorCount + "<br>");
12             document.write("Status Aktif: " + IsActive + "<br>");
13         </script>
14     </head>
15 </html>
16

```

2. output



Deleting Variables

In JavaScript, deleting variables is uncommon. However, in programs where careful memory management is crucial, removing variables is important to ensure more efficient memory usage. This can be achieved using the **delete** keyword.

Example:

```

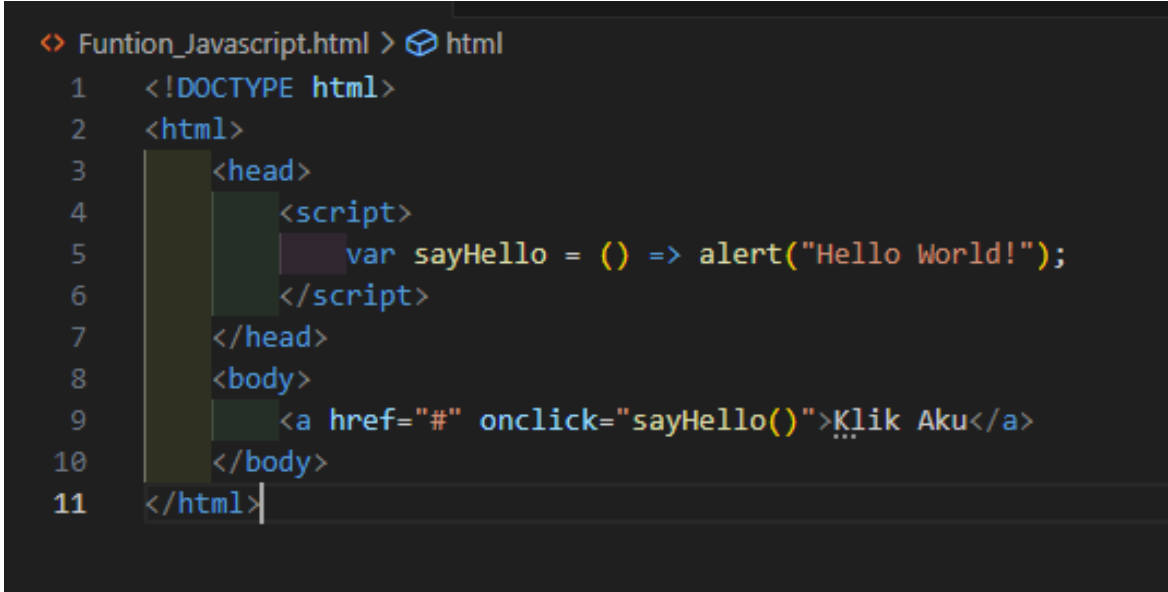
var bookTitle = "Learn Javascript Programming";
delete bookTitle;

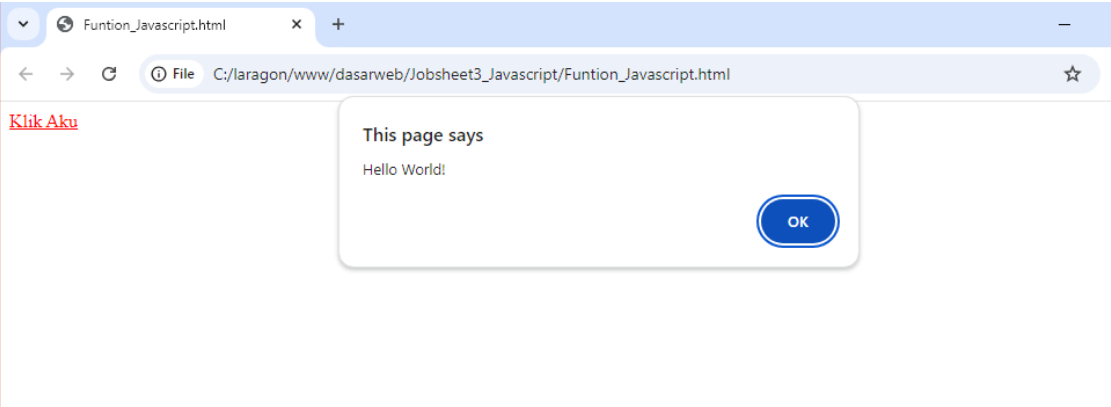
```

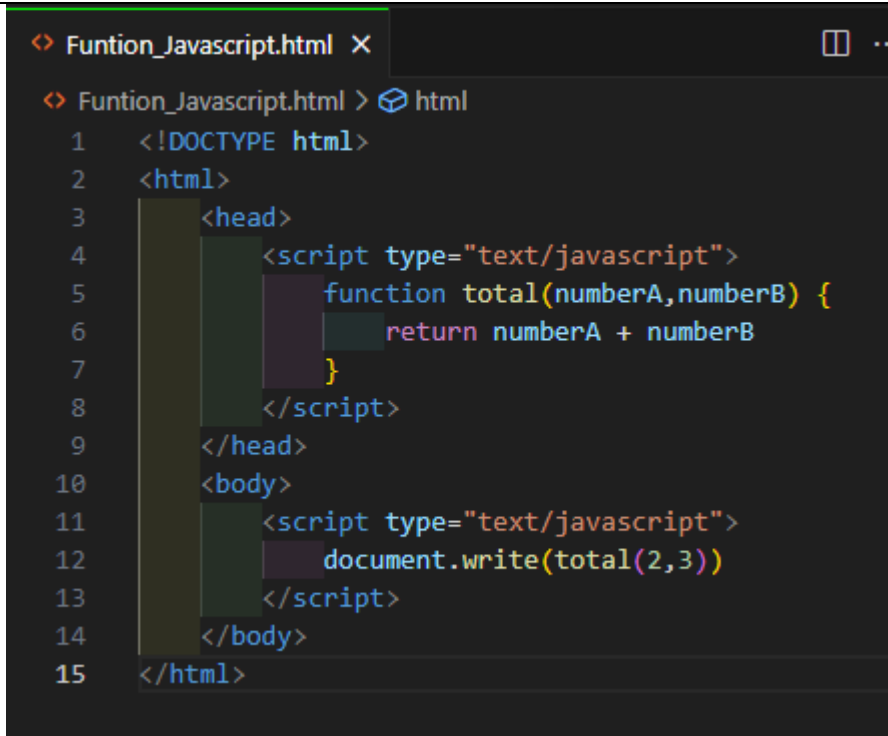
Then the **bookTitle** variable will disappear from memory.

Practical Section 6 : Functions

Functions are sub-programs that can be reused both within the program itself, and in other programs. A function in Javascript is an object. Because it has properties and also *methods*.

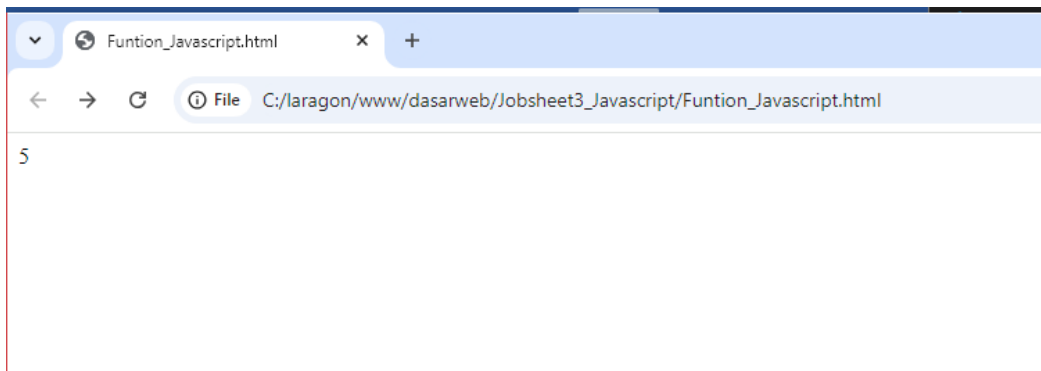
Step	Description
1	Create a new File named <code>function_javascript.html</code> and save it in the project folder
2	How to call a function in Javascript code is usually written with: <code>functionName () ;</code>
3	Type the following program code <pre><!DOCTYPE html> <html> <head> <script> // membuat fungsi var sayHello = () => alert("Hello World!"); </script> </head> <body> <!-- Memanggil fungsi saat link diklik --> Klik Aku! </body> </html></pre>
4	Observe what appears in the browser
5	Record your observations (Question No. 15) a. Input 

	b. Output 
6	A parameter is a variable that stores a value for a process inside a function. How to call a parameter in javascript is: <pre>function kali(a, b){ hasilKali = a * b; console.log("Hasil kali a*b = " + hasilKali); }</pre>
7	Type the following program code <pre><html> <head> <script type="text/javascript"> function total(numberA,numberB) { return numberA + numberB } </script> </head> <body> <script type="text/javascript"> document.write(total(2,3)) </script> </body> </html></pre>
8	Observe what appears in the browser
9	Record your observations (Question No. 16) 1. Input



```
< Funtion_Javascript.html X
< Funtion_Javascript.html > html
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <script type="text/javascript">
5              function total(numberA,numberB) {
6                  return numberA + numberB
7              }
8          </script>
9      </head>
10     <body>
11         <script type="text/javascript">
12             document.write(total(2,3))
13         </script>
14     </body>
15 </html>
```

2. Output



```
5
```

Practical Section 7: Data Types

Data types are the types of data that we can store in variables. There are several types of data in Javascript programming:

- String (text)
- Integer or Number
- Float (number of Fractions)
- Boolean
- Object

Javascript is a *dynamic typing* language, which means that we don't have to write data types when creating variables like in [C](#), [C++](#), [Java](#), etc. which are *static typing*. There are several rules for writing variables in Javascript:

- Variable naming **should not** use numbers in front of it.
example:

```
// wrong
var 123name = "Polinema";

// right
var name123 = "Polinema";
```

- Variable naming **can** use the initial underscore.
example:

```
var _nama = "Polinema";
```

- Variable naming **is recommended** using **camelCase** if it consists of two syllables.
Example:

```
var _fullName = "Polinema";
```

- Variable naming **is recommended** using English
Example:

```
var _postTitle = "Javascript Tutorials";
```

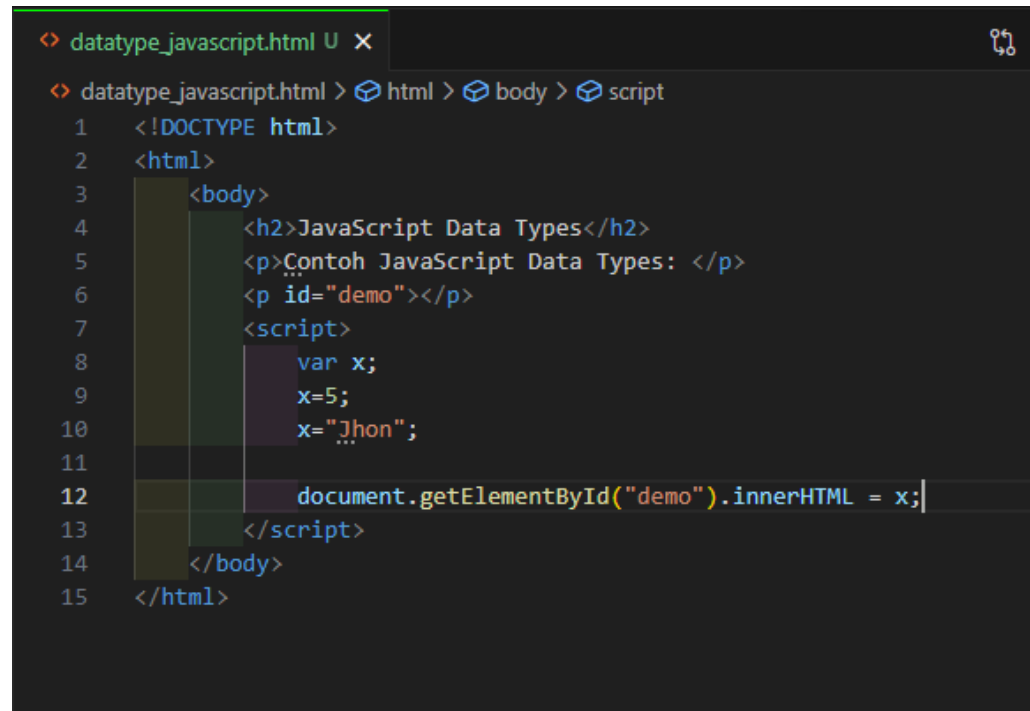
Step	Description
1	Create a new File named datatype_javascript.html and save it in the project folder.
2	Type the following program code <pre><!DOCTYPE html> <html> <body> <h2>JavaScript Data Types</h2> <p>Contoh Javascript Data Types</p> <p id="demo"></p> <script> var x; // Now x is undefined x = 5; // Now x is a Number x = "John"; // Now x is a String document.getElementById("demo").innerHTML = x; </script> </body> </html></pre>
3	Observe what appears in the browser

4

Record your observations

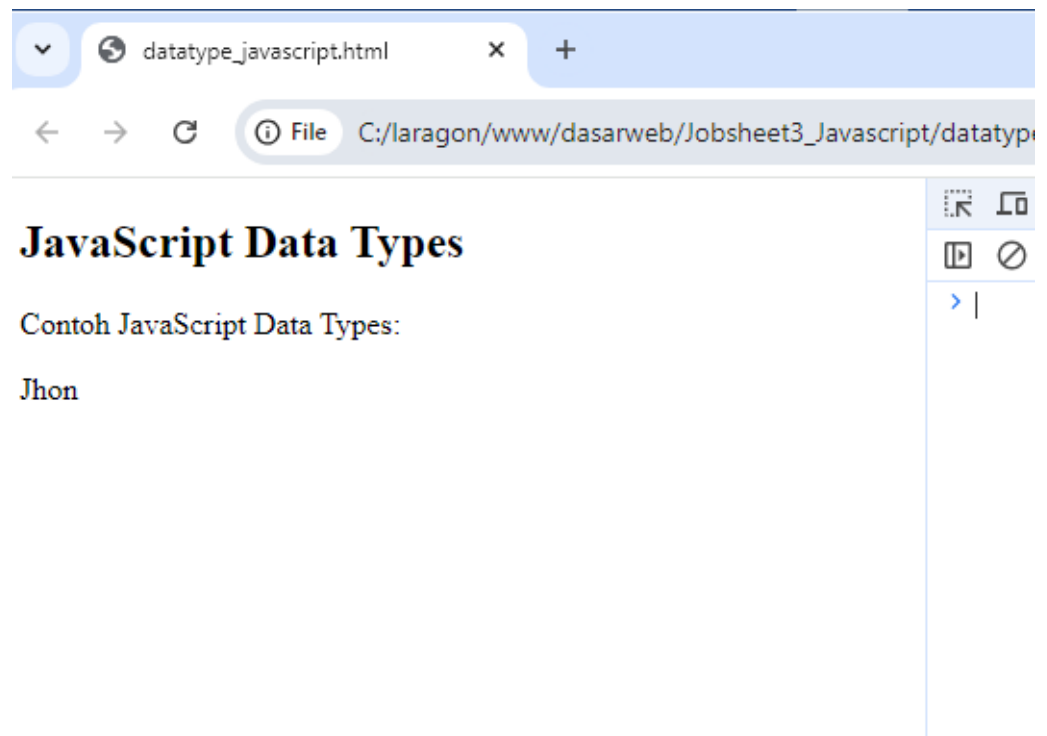
(Question No. 17)

1. Input



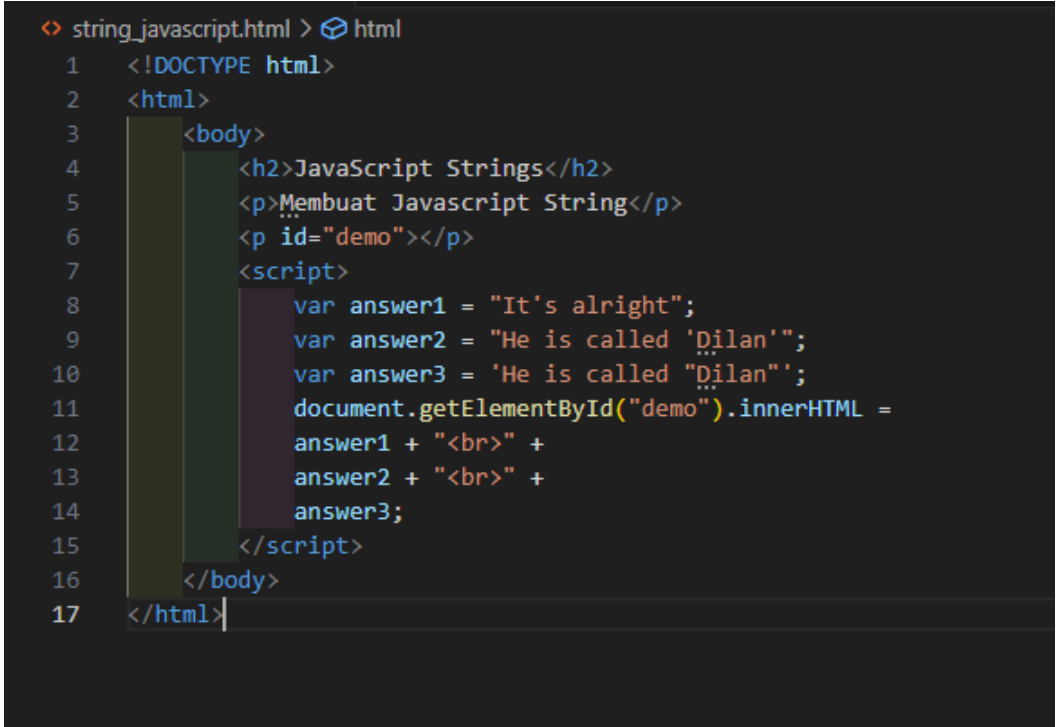
```
<datatype_javascript.html U x>
<datatype_javascript.html > html > body > script
1  <!DOCTYPE html>
2  <html>
3      <body>
4          <h2>JavaScript Data Types</h2>
5          <p>Contoh JavaScript Data Types: </p>
6          <p id="demo"></p>
7          <script>
8              var x;
9              x=5;
10             x="Jhon";
11
12             document.getElementById("demo").innerHTML = x;|
13         </script>
14     </body>
15 </html>
```

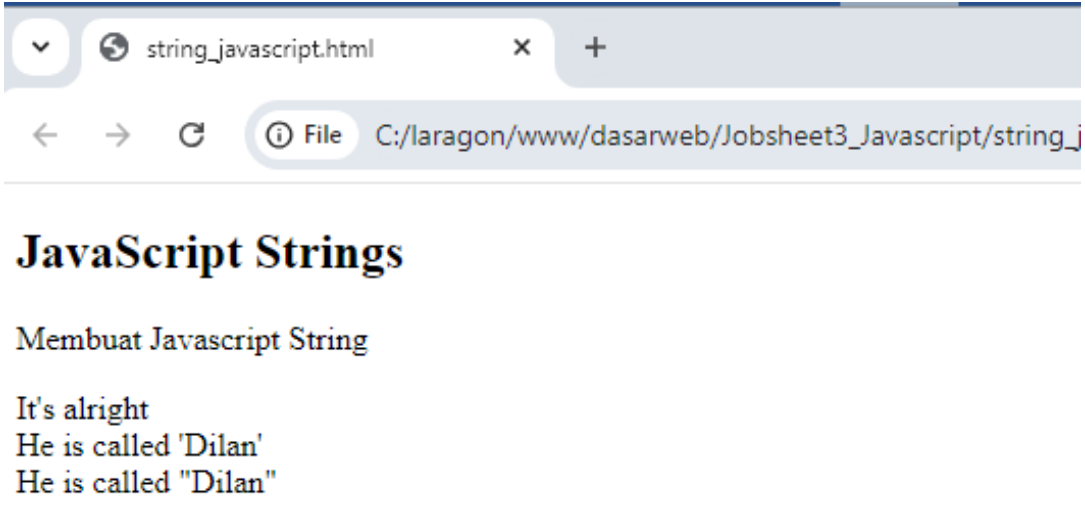
2. Output



5

Type the program below and save it with the `string_javascript.html` name

	<pre> <!DOCTYPE html> <html> <body> <h2>JavaScript Strings</h2> <p>Membuat Javascript String</p> <p id="demo"></p> <script> var answer1 = "It's alright"; var answer2 = "He is called 'Dilan'"; var answer3 = 'He is called "Dilan"'; document.getElementById("demo").innerHTML = answer1 + "
" + answer2 + "
" + answer3; </script> </body> </html> </pre>
6	Observe what appears in the browser
7	Record your observations (Question No. 18) 1. Input 

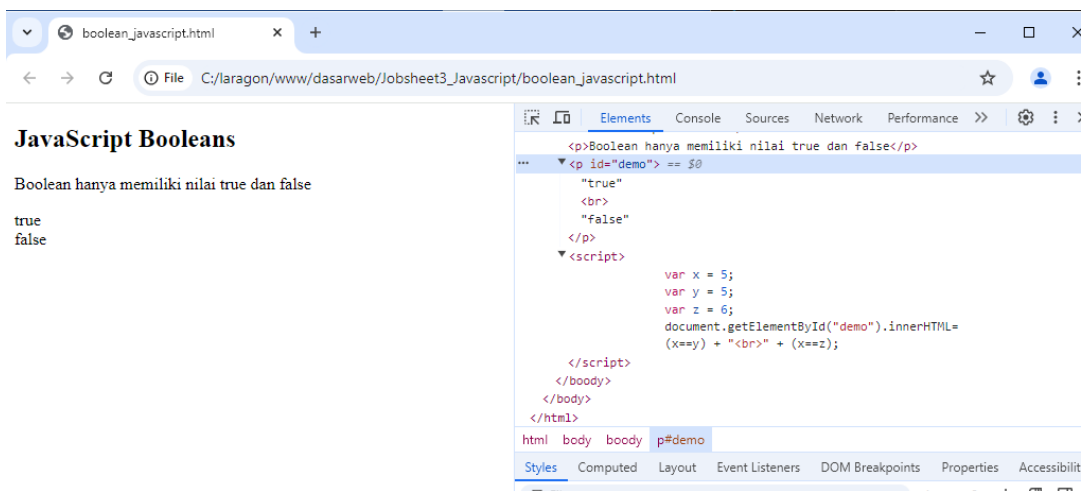
	2. Output 
8	Type the program below and save it with the <code>boolean_javascript.html</code> name <pre> <!DOCTYPE html> <html> <body> <h2>JavaScript Booleans</h2> <p>Booleans hanya memiliki nilai true dan false</p> <p id="demo"></p> <script> var x = 5; var y = 5; var z = 6; document.getElementById("demo").innerHTML = (x == y) + "
" + (x == z); </script> </body> </html> </pre>
9	Observe what appears in the browser
10	Record your observations (Question No. 19) 1. Input

```

<> boolean_javascript.html U x
<> boolean_javascript.html > html > boody > script
1  PE html>
2
3  body>
4  <h2>JavaScript Booleans</h2>
5  <p>Boolean hanya memiliki nilai true dan false</p>
6  <p id="demo"></p>
7
8  <script>
9      var x = 5;
10     var y = 5;
11     var z = 6;
12     document.getElementById("demo").innerHTML=
13     (x==y) + "<br>" + (x==z);
14 </script>
15 body>
16

```

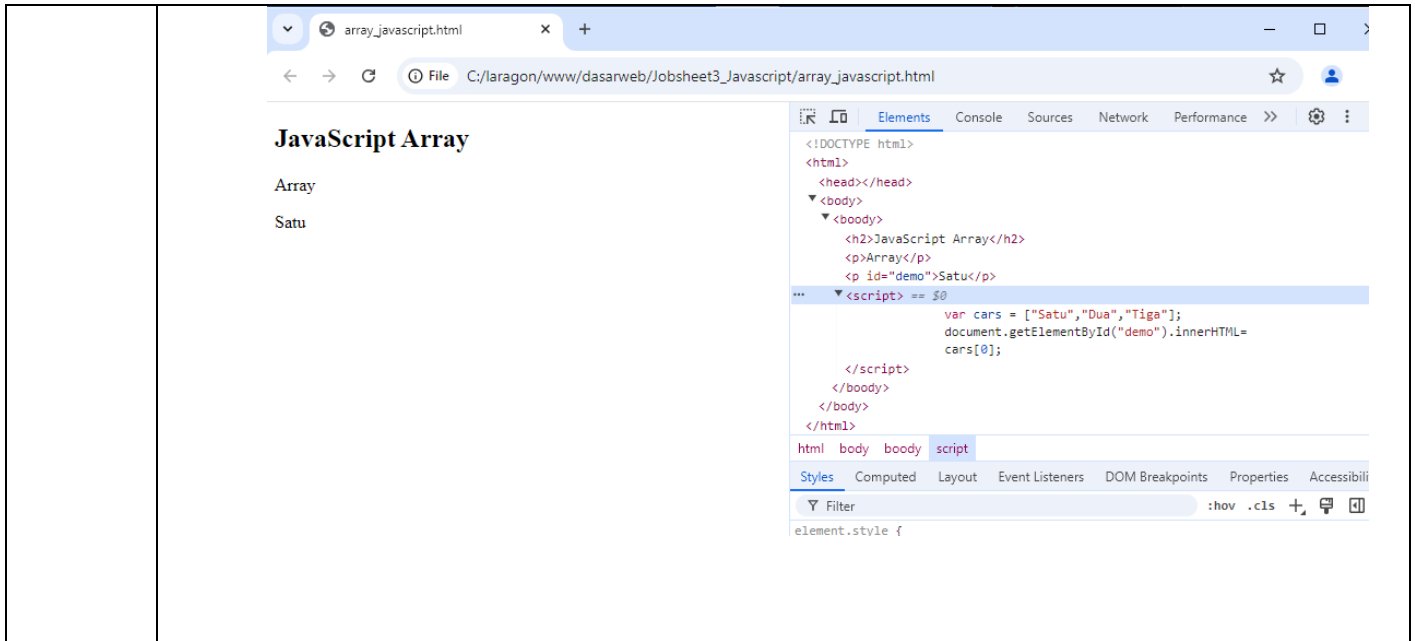
2. Output



11

Type the program below and save it with the **array_javascript.html** name

	<pre> <!DOCTYPE html> <html> <body> <h2>JavaScript Arrays</h2> <p>Array</p> <p id="demo"></p> <script> var cars = ["Satu","Dua","Tiga"]; document.getElementById("demo").innerHTML = cars[0]; </script> </body> </html> </pre>
12	Observe what appears in the browser
13	Record your observations (Question No.20) - Input  <pre> array_javascript.html U X array_javascript.html > html > body > script 1 <!DOCTYPE html> 2 <html> 3 <body> 4 <h2>JavaScript Array</h2> 5 <p>Array</p> 6 <p id="demo"></p> 7 8 <script> 9 var cars = ["Satu","Dua","Tiga"]; 10 document.getElementById("demo").innerHTML= 11 cars[0]; 12 </script> 13 </body> 14 </html> </pre> - Output



Practical Section 8: Operator

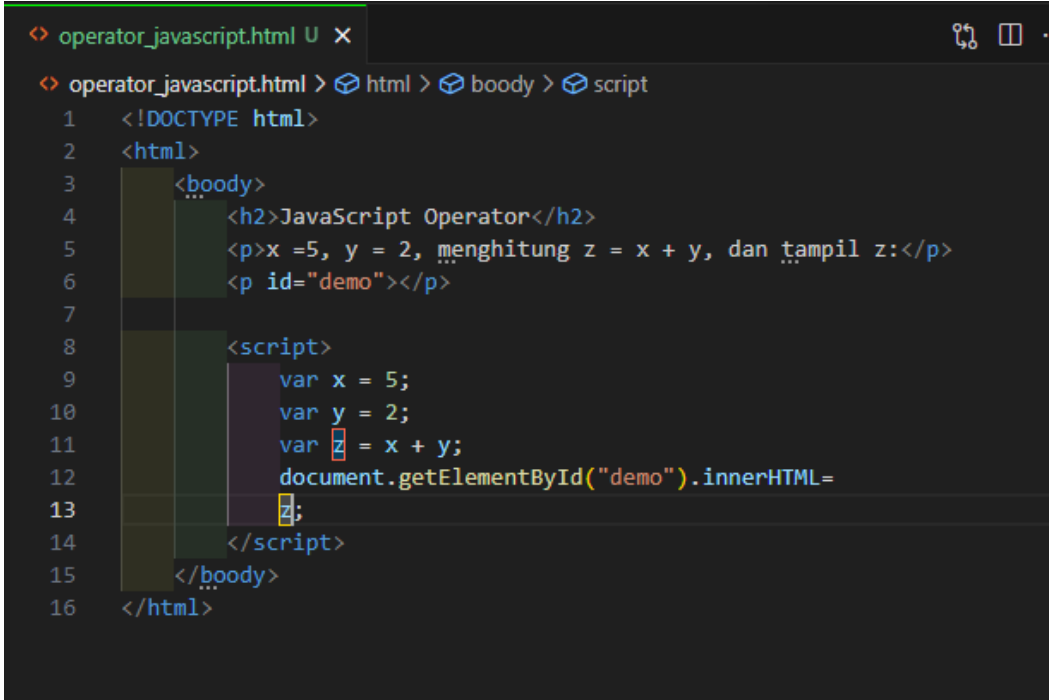
An operator is a symbol used to perform operations on a value and variable. Operators in programming are divided into 6 types:

1. Arithmetic operator;
2. Assignment Operator;
3. relationship or comparison operators;
4. Logic Operators;
5. Bitwise Operator;
6. Ternary Operator;

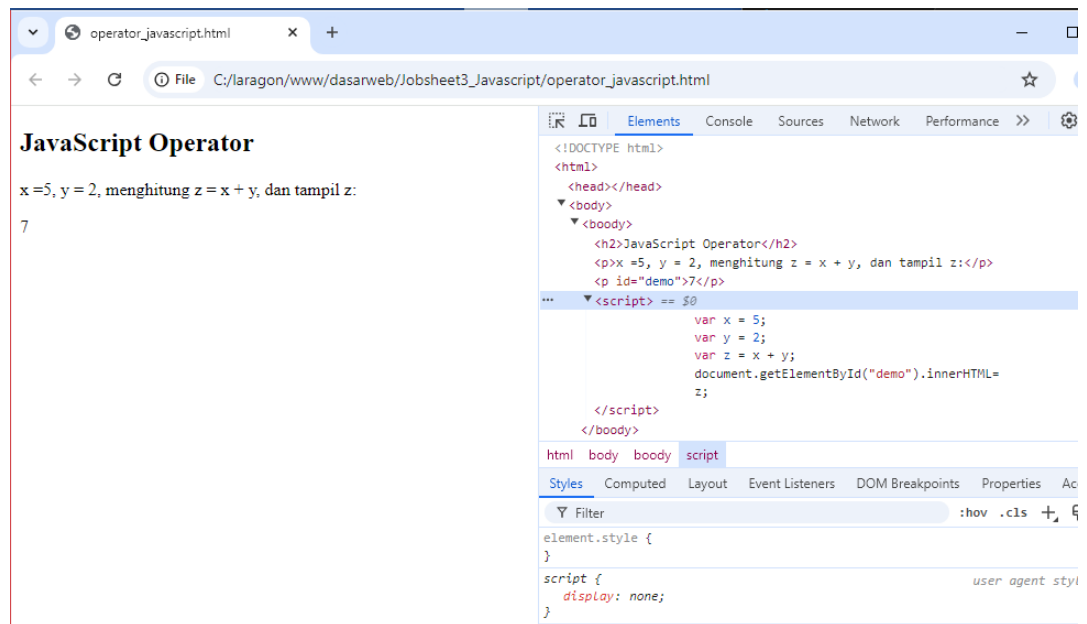
An arithmetic operator is an operator to perform arithmetic operations such as addition, subtraction, division, multiplication, etc. Arithmetic operators consist of:

Operator Name	Symbol
Addition	+
Reduction	-
Multiplication	*
Appointment	**
Division	/
Leftover	%

Step	Description
------	-------------

1	Create a new File named <code>operator_javascript.html</code> and save it in the project folder.
2	Type the program below <pre> <!DOCTYPE html> <html> <body> <h2>JavaScript Operators</h2> <p>x = 5, y = 2, menghitung z = x + y, dan tampil z:</p> <p id="demo"></p> <script> var x = 5; var y = 2; var z = x + y; document.getElementById("demo").innerHTML = z; </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 21) Input  <pre> <? operator_javascript.html U x <? operator_javascript.html > html > boody > script 1 <!DOCTYPE html> 2 <html> 3 <boody> 4 <h2>JavaScript Operator</h2> 5 <p>x =5, y = 2, menghitung z = x + y, dan tampil z:</p> 6 <p id="demo"></p> 7 8 <script> 9 var x = 5; 10 var y = 2; 11 var z = x + y; 12 document.getElementById("demo").innerHTML= 13 z; 14 </script> 15 </boody> 16 </html> </pre>

2. Output



Practical Section 9: Branching

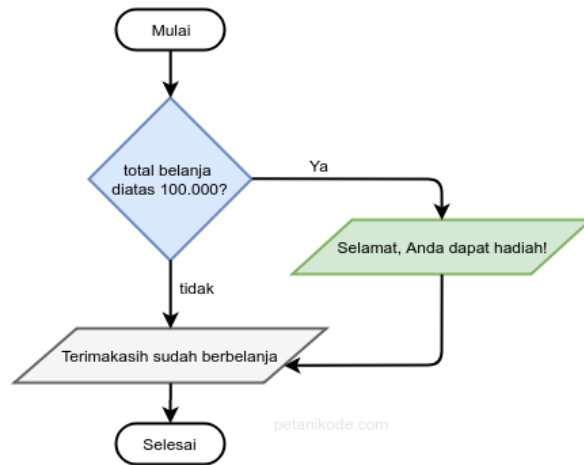
It can be said that branching and looping are one of the core methods in all programming languages worldwide. With branching and looping, a dynamic program can be created instead of a linear and static one. Since JavaScript is a method for client-side web programming, it also has this capability.

Some branching functions include:

- Use `if` to specify a block of code to be executed, if a specified condition is true
- Use `else` to specify a block of code to be executed, if the same condition is false
- Use `else if` to specify a new condition to test, if the first condition is false
- Use `switch` to specify many alternative blocks of code to be executed

❖ if Branching

if branching is a structure that only has one block of choice when the condition is true. Take a look at the following flowchart:



“If the total purchase is greater than Rp 100,000, then display the message: Congratulations, you won a prize.”

What if it is below Rp 100,000?

Yes, the message is not displayed.

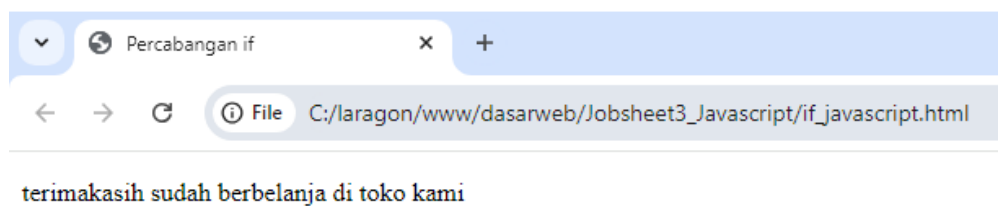
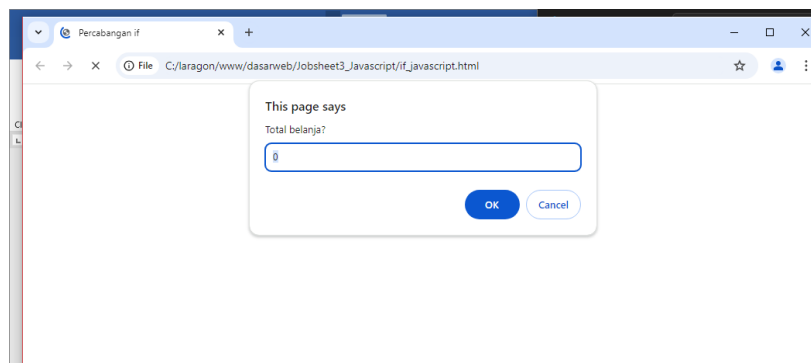
Step	Description
1	Create a new File named if_javascript.html and save it in the project folder
2	Type the program below <pre> <!DOCTYPE html> <html lang="en"> <head> <title>Percabangan if</title> </head> <body> <script> var totalBelanja = prompt("Total belanja?", 0); if(totalBelanja > 30000){ document.write("<h2>Selamat Anda dapat hadiah</h2>"); } document.write("<p>Terimakasih sudah berbelanja di toko kami</p>"); </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 22) 1. input

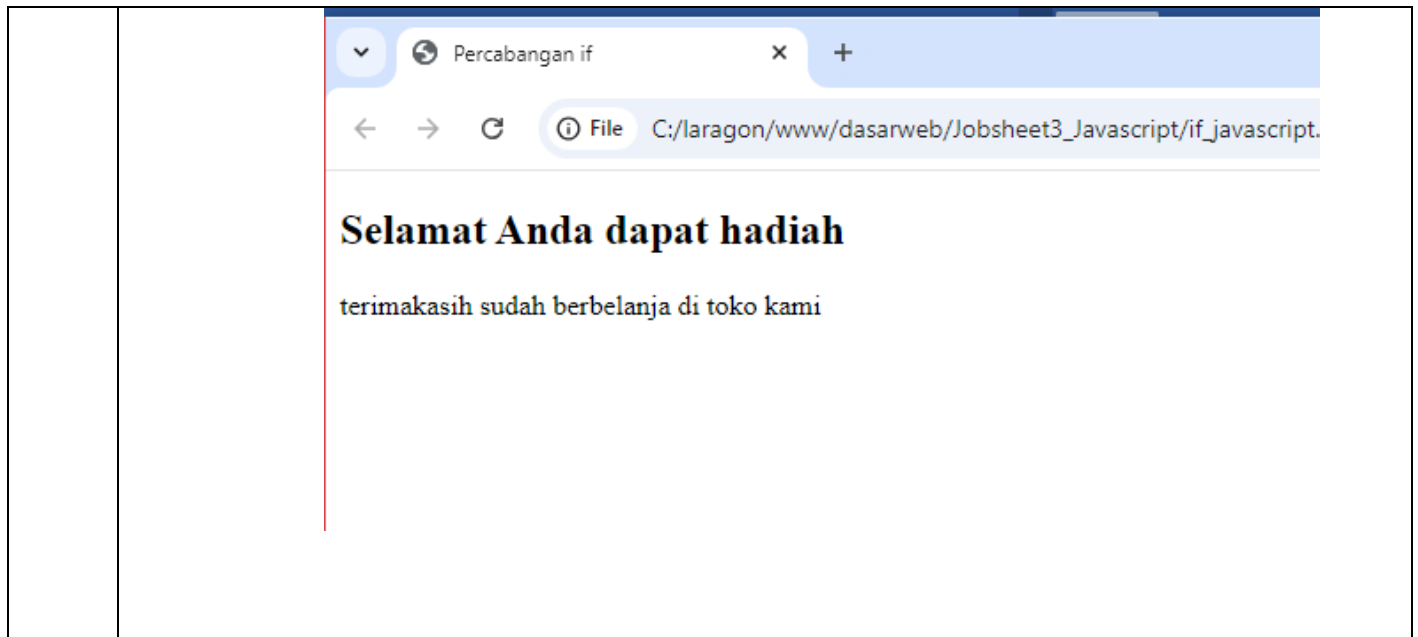
```

if_javascript.html U x
if_javascript.html > html > body > script
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>Percabangan if</title>
5 </head>
6 <body>
7 <script>
8     var totalBelanja = prompt ("Total belanja?", 0);
9
10    if (totalBelanja > 30000) {
11        document.write("<h2>Selamat Anda dapat hadiah</h2>");
12    }
13    document.write("<p>terimakasih sudah berbelanja di toko kami</p>")
14 </script>
15 </body>
16 </html>

```

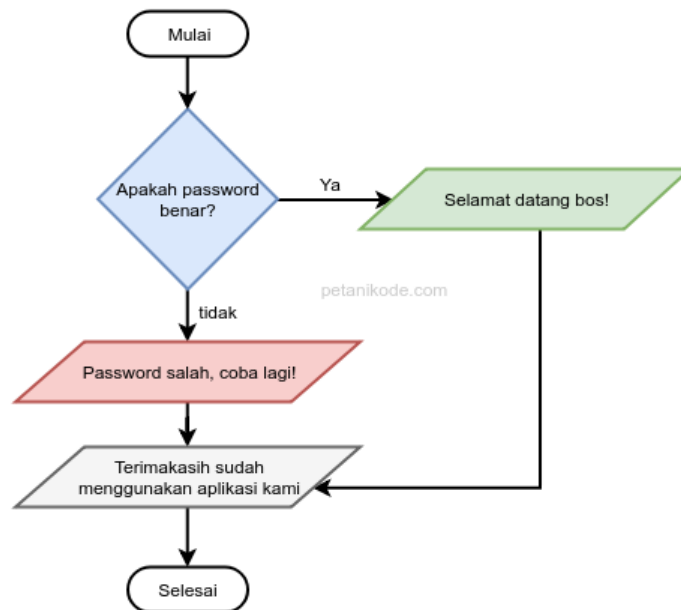
2. Output





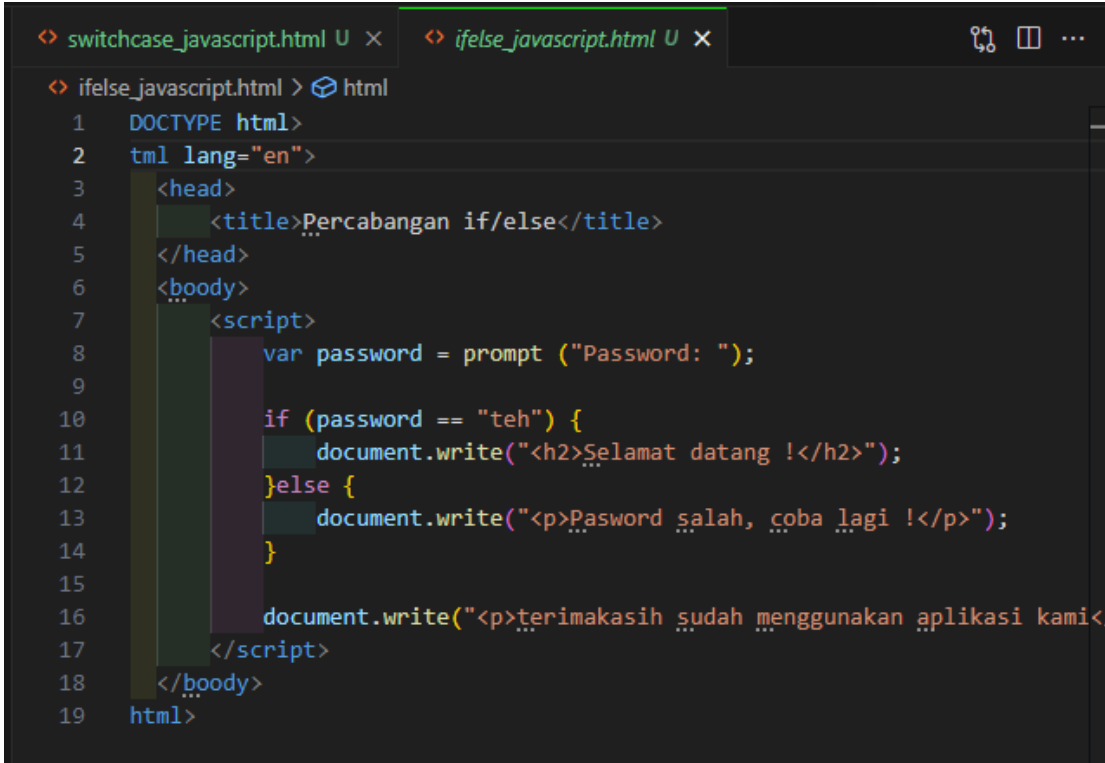
❖ if/else Branching

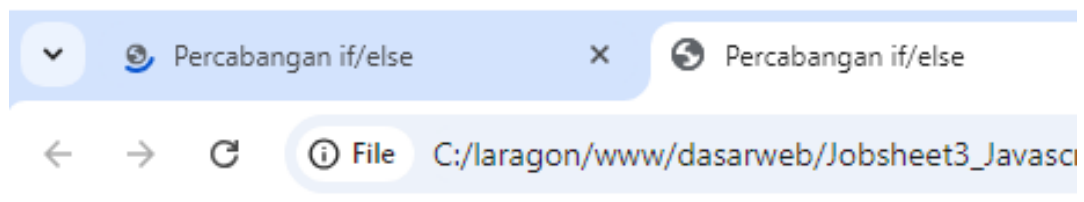
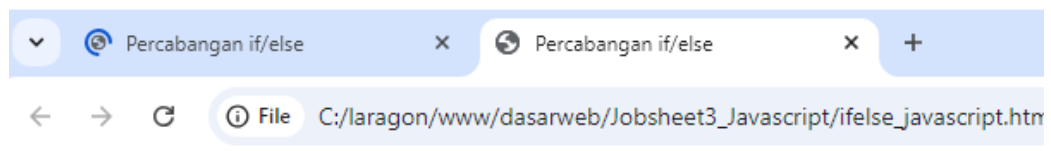
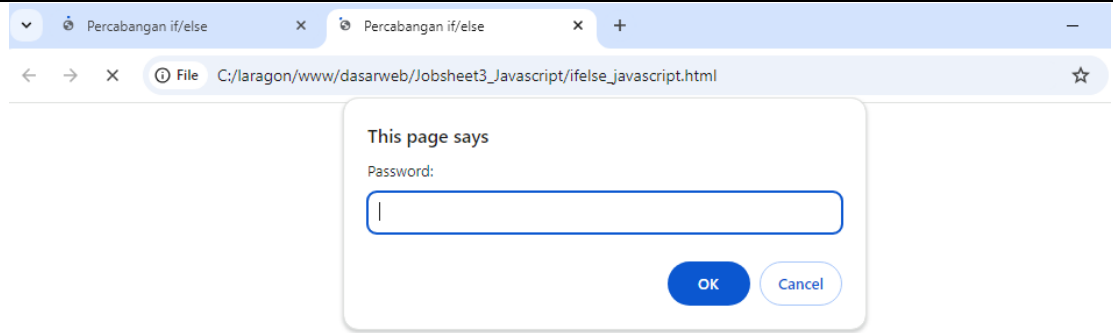
If/Else Branching is a structure that has **two blocks of choices**. The first choice is for when the **condition is true**, and the second choice is for when the **condition is false (else)**. Take a look at this flowchart:



This is a flowchart for checking the password. If the password is correct, the message in the green block will be displayed: **"Welcome, boss!"** However, if it is incorrect, the message in the red block will be shown: **"Incorrect password, please try again!"**

Step	Description
------	-------------

1	Create a new File named <code>ifelse_javascript.html</code> and save it in the project folder.
2	Type the program below <pre> <!DOCTYPE html> <html> <head> <title>Percabangan if/else</title> </head> <body> <script> var password = prompt("Password:"); if(password == "teh"){ document.write("<h2>Selamat datang !</h2>"); } else { document.write("<p>Password salah, coba lagi!</p>"); } document.write("<p>Terima kasih sudah menggunakan aplikasi ini!</p>"); </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 23) - input  - output



--	--

❖ switch/case Branching

switch/case branching is an alternative form of the **if/else/if** branching structure. In a **switch/case** statement, instead of evaluating multiple if conditions, the program evaluates the value of a variable or expression and compares it against multiple possible cases. Each case represents a potential value, and when a match is found, the corresponding block of code is executed. If no case matches, the **default** case is executed (if provided), similar to the **else** block in **if/else** statements.

The **switch/case** structure can make code more readable and organized, especially when dealing with multiple conditions based on a single variable. The structure looks like this:

```
switch(variabel){
    case <value>:
        // blok kode
        break;
    case <value>:
        // blok kode
        break;
    default:
        // blok kode
}
```

Step	Description
1	Create a new File named switchcase_javascript.html and save it in the project folder
2	Type the program below

	<pre> <!DOCTYPE html> <html> <head> <title>Percabangan switch/case</title> </head> <body> <script> var jawab = prompt("Kamu beruntung! Silahakn pilih hadiahmu dengan memasukan angka 1 sampai 5"); var hadiah = ""; switch(jawab){ case "1": hadiah = "Tisu"; break; case "2": hadiah = "1 Kotak Kopi"; break; case "3": hadiah = "Sticker"; break; case "4": hadiah = "Minyak Goreng"; break; case "5": hadiah = "Uang Rp 50.000"; break; default: document.write("<p>Opps! anda salah pilih</p>"); } if(hadiah == ""){ document.write("<p>Kamu gagal mendapat hadiah</p>"); } else { document.write("<h2>Selamat kamu mendapatkan " + hadiah + "</h2>"); } </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 24) 1. input

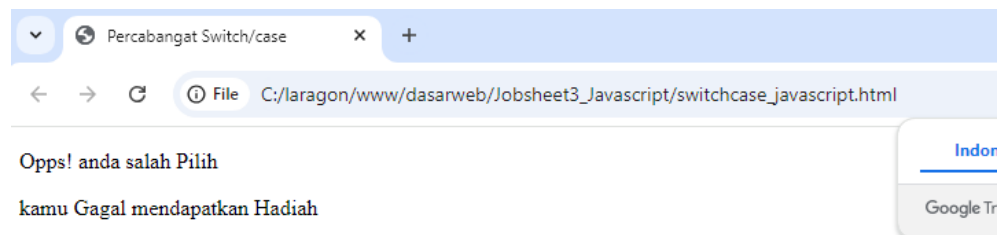
```

switchcase_javascript.html U x
switchcase_javascript.html > html > boody > script
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Percabangat Switch/case</title>
5      </head>
6      <boody>
7          <script>
8              var jawab = prompt("Kamu Beruntung ! Silahkan Pilih hadiah mu d
9              var hadiah = "";
10             switch (jawab) {
11                 case "1":
12                     hadiah = "Tissue";
13                     break;
14                 case "2":
15                     hadiah = "1 Kotak Kopi";
16                     break;
17                 case "3":
18                     hadiah = "Sticker";
19                     break;
20                 case "4":
21                     hadiah = "Minyak Goreng";
22                     break;
23                 case "5":
24                     hadiah = "Uang Rp. 50.000";
25                     break;
26                 default:
27             }
28             document.write("<p> Opps! anda salah Pilih</p>");
29             if (hadiah === "") {
30                 document.write("<p>kamu Gagal mendapatkan Hadiah</p>");
31             } else {
32                 document.write("<h2> Selamat kamu mendapatkan" + hadiah +
33             }
34

```

2. output

Selamat kamu mendapatkan Uang Rp. 50.000



❖ Nested Branching

Nested Branching refers to a condition where one branching statement (such as **if**, **else**, **switch**, etc.) is placed inside another branching statement. This allows for more complex decision-making processes where multiple conditions need to be evaluated at different levels. In nested branching, the outcome of one condition can depend on the result of another, providing more fine-grained control over the program flow.

For example, you can nest an **if** statement inside another **if** statement to first check one condition and then, based on that, check a second condition within the first block.

Step	Description
1	Create a new File named <code>nestedif_javascript.html</code> and save it in the project folder
2	Type the program below <pre><!DOCTYPE html> <html> <head> <title>Percabangan Ternary</title> </head> <body> <script> var username = prompt("Username:"); var password = prompt("Password:"); if(username == "mahasiswa"){ if(password == "kopi"){ document.write("<h2>Selamat datang </h2>"); } else { document.write("<p>Password salah, coba lagi!</p>"); } } else { document.write("<p>Anda tidak terdaftar!</p>"); } </script> </body> </html></pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 25) 1. Input

```
<> switchcase_javascript.html U <> nestedif_javascript.html U x
<> nestedif_javascript.html > html
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Percabangan Ternery</title>
5   </head>
6   <body>
7     <script>
8       var username = prompt("username:");
9       var password = prompt("Password:");
10
11       if(username=="Mahasiswa"){
12         if (password === "kopi") {
13           document.write("<h2>Selamat Datang</h2>");
14         } else {
15           document.write("<p> Password salah, coba lagi!</p>");
16         }
17       } else {
18         document.write("<p> Anda tidak terdaftar!</p>");
19       }
20     </script>
21   </body>
22 </html>
```

2. Output



Practical Section 10: Loops

Loops help us execute code repeatedly, as many times as we want. There are five types of loops in JavaScript. Generally, these loops are categorized into two types: counted loops and uncounted loops.

The difference is as follows:

- ✓ **Counted Loops** are loops where the number of iterations is **known** and **definite**.
- ✓ **Uncounted Loops**, on the other hand, are loops where the number of iterations is **not predetermined**.

The loops that fall under **Counted Loops** are:

1. For Loop
2. Foreach Loop
3. Repeat Loop

The loops that fall under **Uncounted Loops** are:

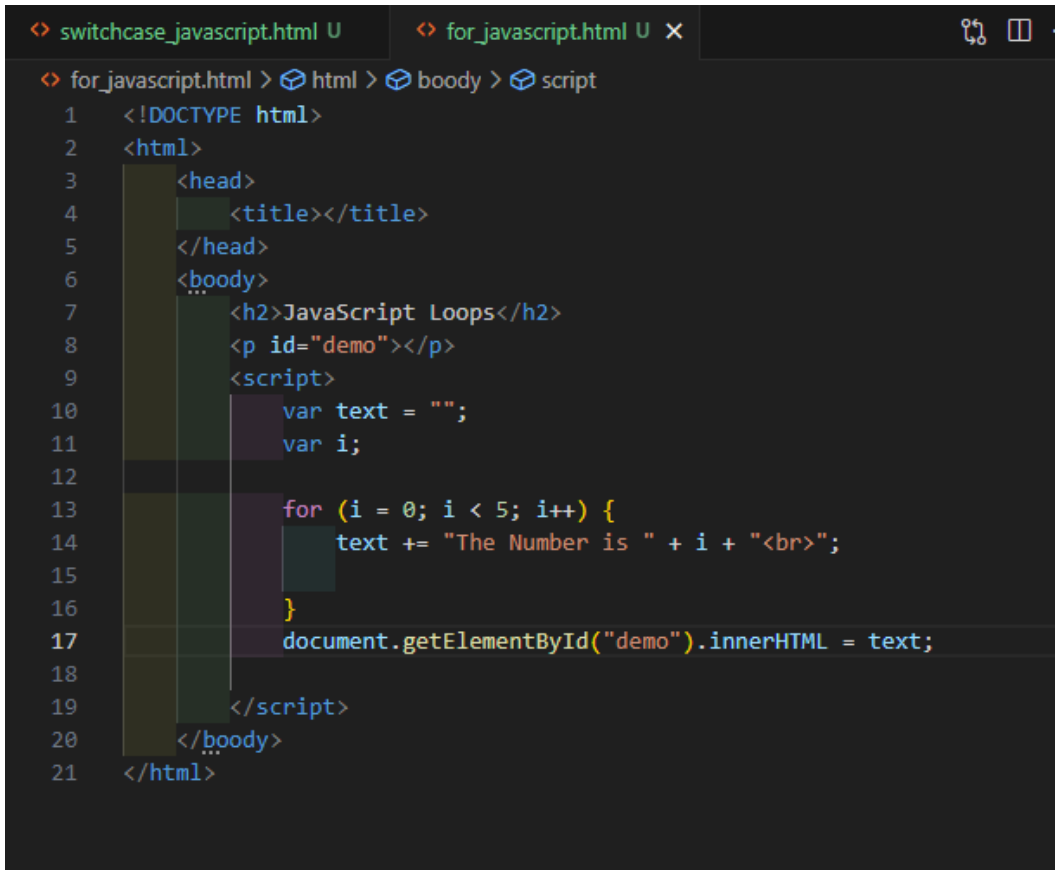
1. While Loop
2. Do/While Loop

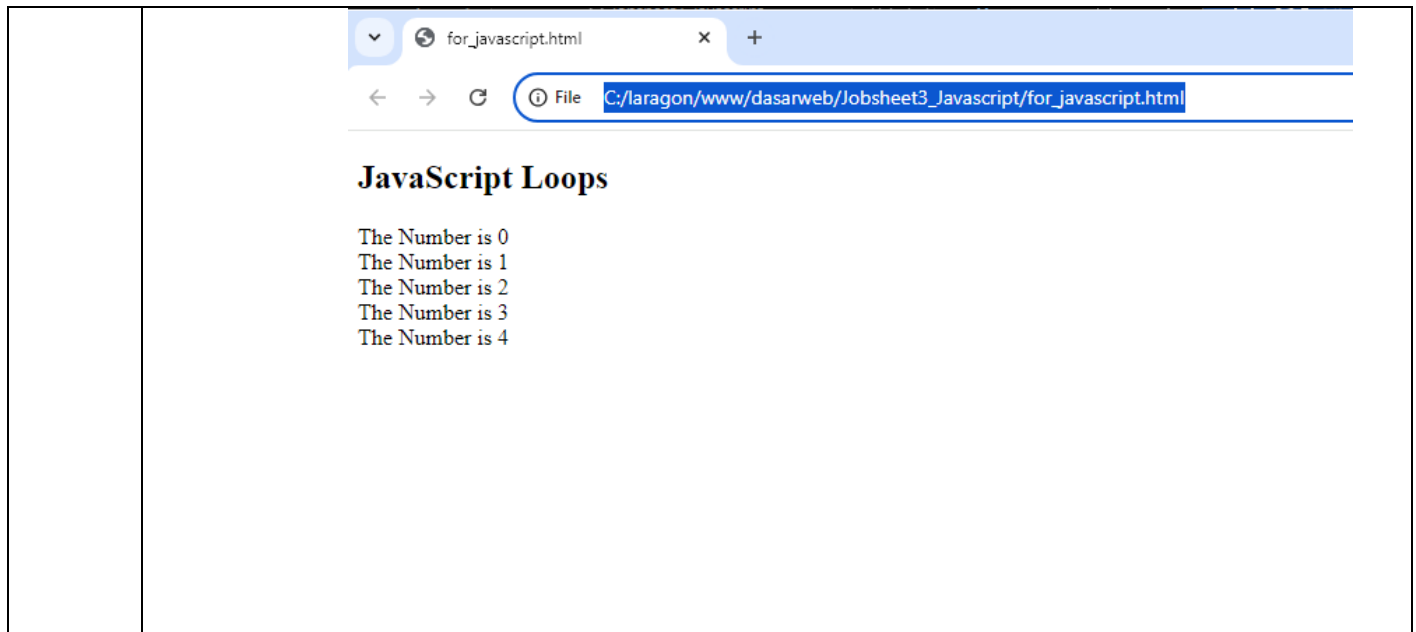
❖ For loops in Javascript

A for loop is a loop that is included in a *counted loop*, because it is clear how many times it will repeat. It looks like this:

```
for(let i = 0; i < 10; i++){  
    document.write("<p>Perulangan ke-" + i + "</p>")  
}
```

Step	Description
1	Create a new File named <code>for_javascript.html</code> and save it in the project folder
2	Type the program below

	<pre> <!DOCTYPE html> <html> <body> <h2>JavaScript Loops</h2> <p id="demo"></p> <script> var text = ""; var i; for (i = 0; i < 5; i++) { text += "The number is " + i + "
"; } document.getElementById("demo").innerHTML = text; </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 26) 1. Inpu  2. Output

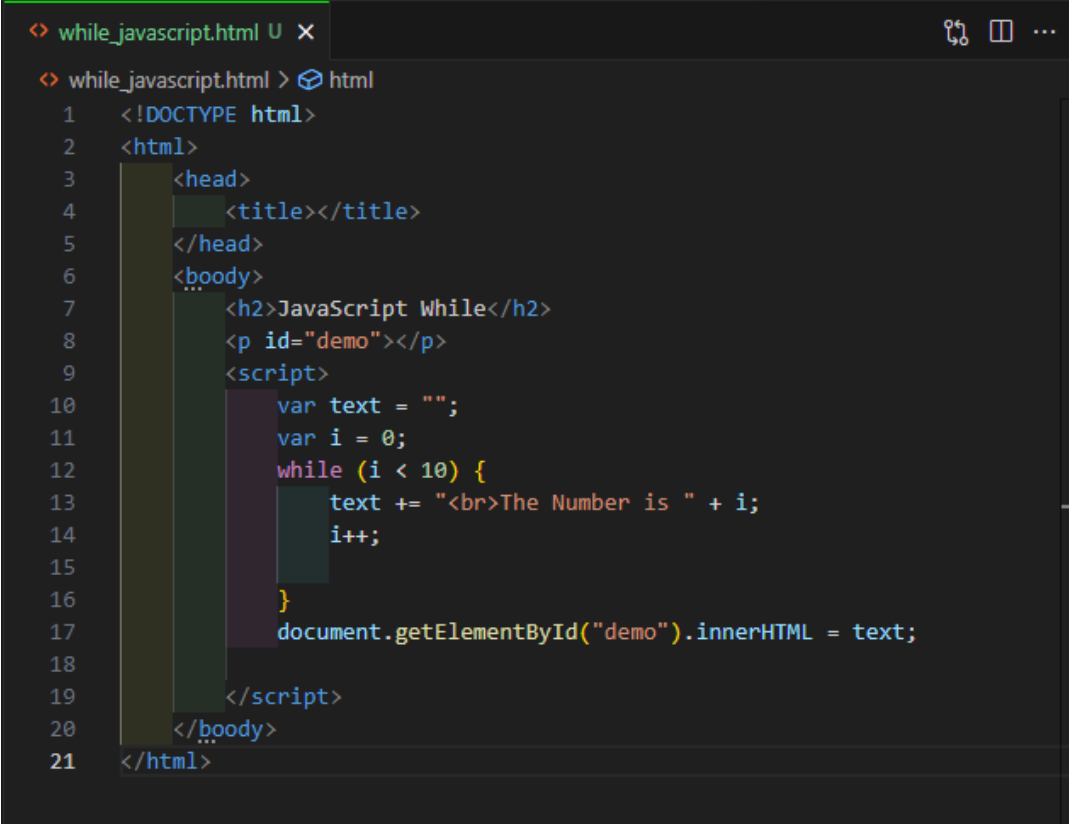


❖ While loops in Javascript

The while loop is categorized as an uncounted loop. However, the while loop can also function as a counted loop by including a counter within it.

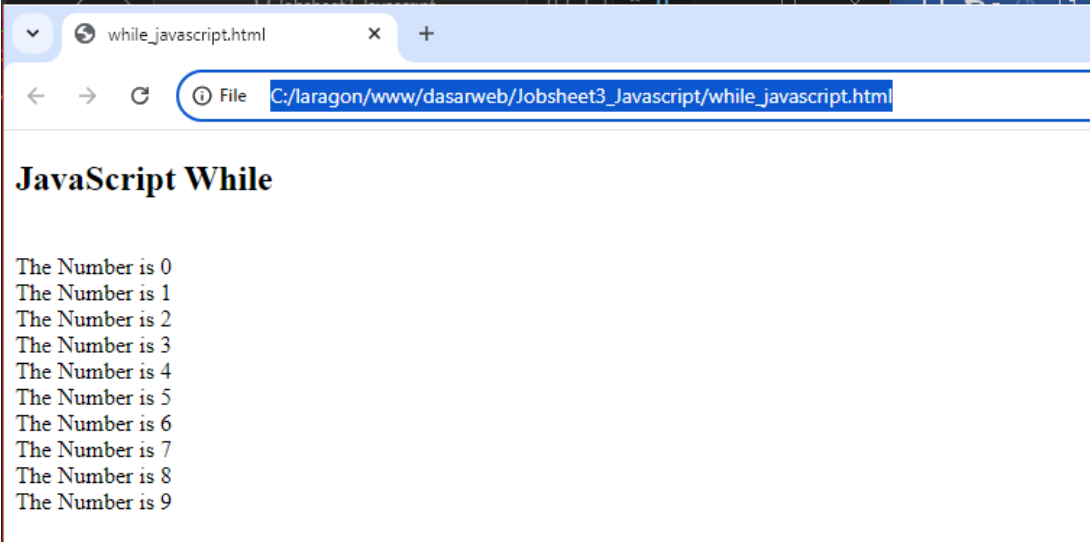
Step	Description
1	Create a new File named <code>while_javascript.html</code> and save it in the project folder
2	Type the program below <pre> <!DOCTYPE html> <html> <body> <h2>JavaScript while</h2> <p id="demo"></p> <script> var text = ""; var i = 0; while (i < 10) { text += "
The number is " + i; i++; } document.getElementById("demo").innerHTML = text; </script> </body> </html> </pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 27)

1. Input



```
<? while_javascript.html U x
<? while_javascript.html > html
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title></title>
5      </head>
6      <body>
7          <h2>JavaScript While</h2>
8          <p id="demo"></p>
9          <script>
10             var text = "";
11             var i = 0;
12             while (i < 10) {
13                 text += "<br>The Number is " + i;
14                 i++;
15             }
16             document.getElementById("demo").innerHTML = text;
17         </script>
18     </body>
19 </html>
```

2. Output



JavaScript While

The Number is 0
The Number is 1
The Number is 2
The Number is 3
The Number is 4
The Number is 5
The Number is 6
The Number is 7
The Number is 8
The Number is 9

❖ Do/While Loops in Javascript

The **do/while loop** is a variation of the while loop in JavaScript. The main difference between them is that the do/while loop will always execute the code inside the loop **at least once**, regardless of whether the

condition is true or false. This is because the condition is evaluated **after** the code block is executed, not before, as in the standard while loop.:

```
do {
    // blok kode yang akan diulang
} while (<kondisi>);
```

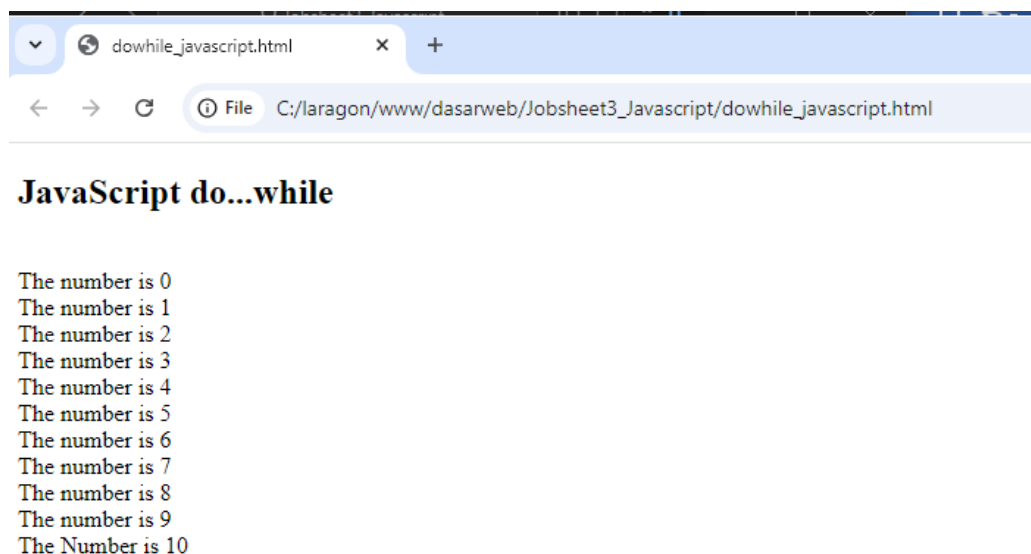
Key Characteristics:

- The code inside the do block runs first, and then the condition is checked.
- If the condition is true, the loop repeats; if false, the loop stops.
- This type of loop ensures that the code inside the loop executes at least once, even if the condition is false from the beginning.

Step	Description
1	Create a new File named <code>dowhile_javascript.html</code> and save it in the project folder
2	Type the program below <pre><!DOCTYPE html> <html> <body> <h2>JavaScript do ... while</h2> <p id="demo"></p> <script> var text = "" var i = 0; do { text += "
The number is " + i; i++; } while (i < 10); document.getElementById("demo").innerHTML = text; </script> </body> </html></pre>
3	Observe what appears in the browser
4	Record your observations (Question No. 28) 1. Input

```
dowhile_javascript.html U x
dowhile_javascript.html > html > boody > script
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title></title>
5      </head>
6      <boody>
7          <h2>JavaScript do...while</h2>
8          <p id="demo"></p>
9          <script>
10             var text = "";
11             var i = 0;
12
13             do{
14                 text += "<br>The number is " + i;
15                 i++;
16             }
17             while (i < 10) {
18                 text += "<br>The Number is " + i;
19                 i++;
20             }
21             document.getElementById("demo").innerHTML = text;
22         </script>
23     </boody>
24 </html>
```

2. Output



Reference:

- 1) Jason Beaird, The principles of Beautiful Web Design
- 2) Rian Ariona, Learn HTML and CSS (Fundamental Tutorial in Learning HTML and CSS)
- 3) Adi Hadisaputra, HTML and CSS Fundamentals from the Roots to the Leaves of John Duckett, HTML and CSS design and build websites